Hands-On sessions: Linking motifs to variant effects

ISMB-ECCB 2025: *Tutorial IP2 (MPRAs and IGVF Resource)*

Kilian Salomon
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July 20, 2025

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Berlin Institute of Health @Charité

Hands-on: Variant Effects and Motifs

https://shorturl.at/BpGa5

https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial/tree/main/06_variant_effects_and_motifs

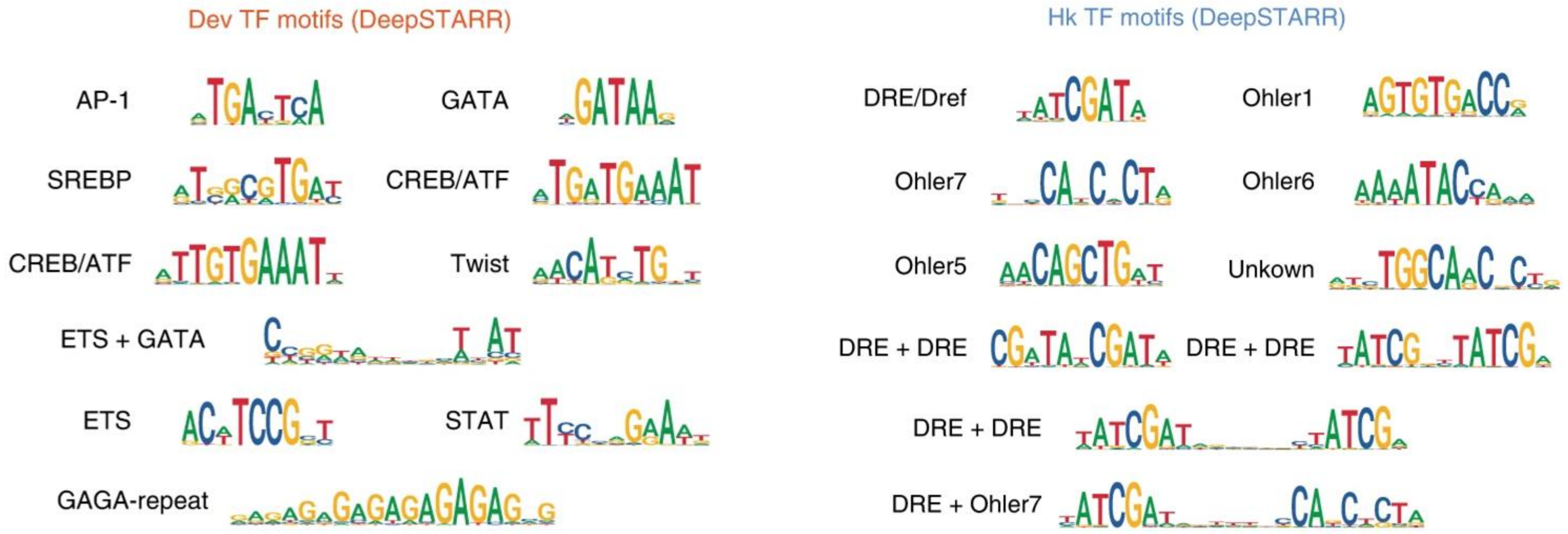
1. Introduction: Variants Tested with MPRA

- **Data sources:** MPRAsnakeflow results, Variant map, BCalm variant effects
- **Steps:**
  1. Normalize counts
  2. Run BCalm using normalized counts and variant map
  3. Identify significant variants

2. Transcription Factor Binding Motif Identification

- **Databases:**
  - JASPAR2022
  - HOCOMOCO v13
- **Tools:**
  - HOMER
  - FIMO (MEME suite)
- **Workflow:**
  - Run FIMO on test data
  - Merge TFBS information with variant metadata
  - Assess if TFBS are gained or lost due to variants
  - Increase confidence in TFBS calls (e.g., by filtering for cell-type expression)
  - Filter for variant-overlapping TFBS
  - Visualize the proportion of variants overlapping TFBS

Transcription factor motifs and their relation to cell-type and allelic differences



de Almeida, B.P., Reiter, F., Pagani, M. *et al.* DeepSTARR predicts enhancer activity from DNA sequence and enables the de novo design of synthetic enhancers. *Nat Genet* 54, 613–624 (2022).

Technical Note on Transcription Factor Motif Discovery from Importance Scores (TF-MoDISco) version 0.5.6.5

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Abstract

Deep Learning has recently gained popularity in genomic sequence modeling tasks [1, 4, 12], but interpretation of these models remains challenging. Of particular interest is understanding the recurring patterns learned by the model. Existing motif discovery methods for neural networks visualize individual convolutional filters [5, 1, 4], but these methods do not account for the fact that deep neural networks learn distributed representations where multiple neurons cooperate to describe a single pattern. Thus, the patterns recognized by individual filters are often found to be partially redundant with each other, hampering interpretability and making it difficult to obtain a non-redundant set of predictive motifs learned by the neural network. In this technical note, we describe TF-MoDISco (Transcription Factor Motif Discovery from Importance Scores), a novel algorithm that leverages per-base importance scores to simultaneously incorporate information from all neurons in the network and generate high-quality, consolidated, non-redundant motifs. This technical note focuses on version 0.5.6.5. The implementation is available at (`https://github.com/kundajelab/tfmodisco/tree/v0.5.6.5`).

Obtaining genetics insights from deep learning via explainable artificial intelligence

Gherman Novakovsky [1,2,7], Nick Dexter [3,4,7], Maxwell W. Libbrecht [4,8]✉, Wyeth W. Wasserman [1,8]✉ and Sara Mostafavi [5,6,8]✉

From local propagation results to a global interpretation. To generalize from per-sequence attribution maps that are produced by propagation-based methods to reveal a global understanding of important motifs, the results across many input examples need to be aggregated. TFMoDisco is one such approach that is specifically designed for DNA input sequences and relies on clustering across attribution maps to identify globally important sequence motifs[63]. As one example, TFMoDisco

Scope of today's workshop

Goals

- Investigating variant-based dataset
- Get to know a motif database
- Learn about workflow of finding and interpreting motifs
- Functional validation in living systems is a cycle
- Glimpse on the complexity of TFBS grammar

Questions we will cover:

- Which known transcription factors can be related to the variant effects?

Open the Notebook of this Hands-on tutorial:

Github repository: https://github.com/kircherlab/ISMB-2025_IGVF-MPRA-Tutorial

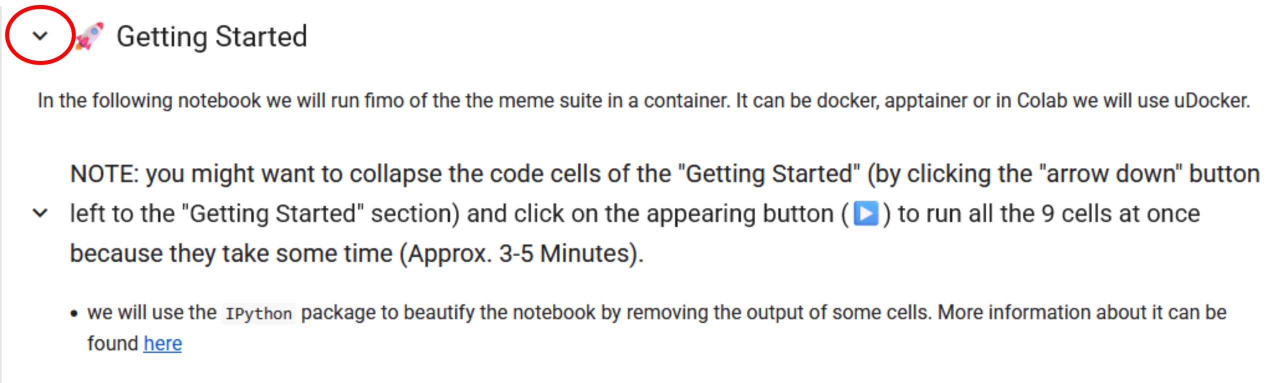
- Go to: 06_variant_effects_and_motifs/from_variant_effects_to_motifs.ipynb
- Click Open in Colab

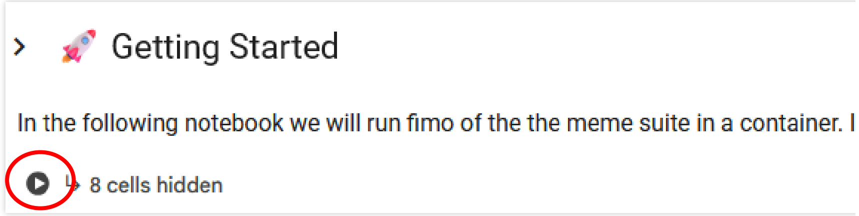
Or use the short URL: https://shorturl.at/BpGa5

Getting started

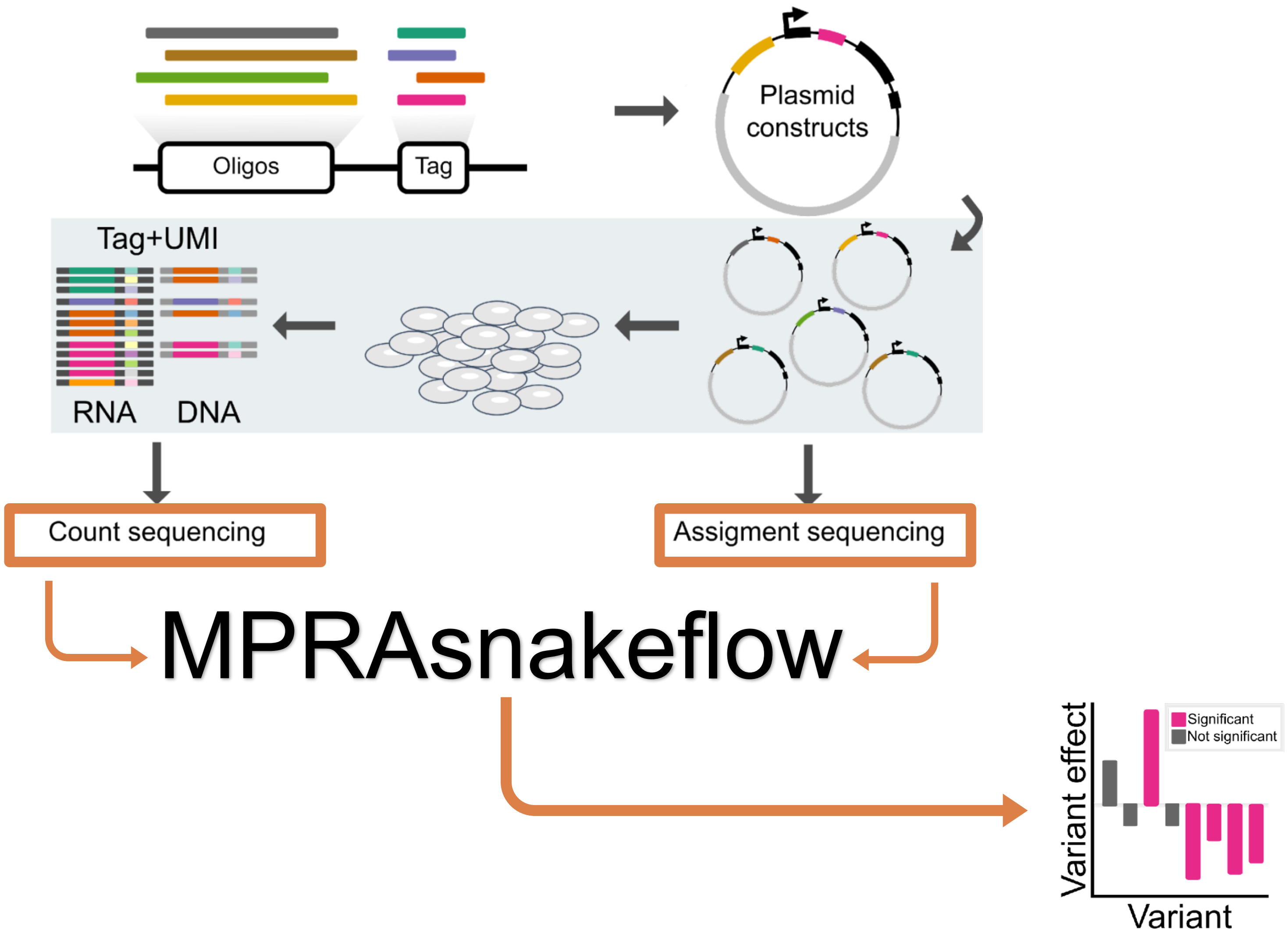
The installation of the required packages takes some time so let‘s start it first

- Please collapse the „Getting Started“ section and run all cells at once
  - By clicking on the marked arrow
  - Followed by clicking the run button

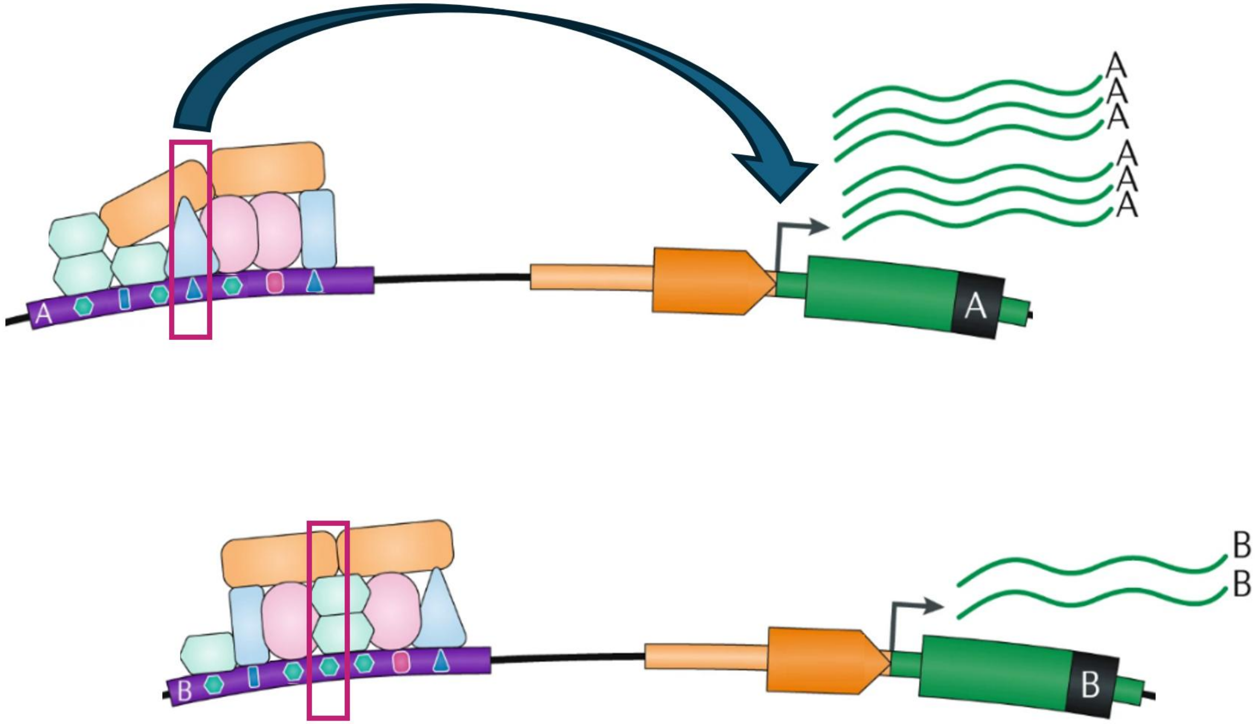




Recap: Massively Parallel Reporter Assay (MPRA)



Recap: Measured Variant Effect



Mod. from Gasperini et al, *Nat Reviews Genetics* 2020

Alternative and Reference allele upstream

Variant effect on transcriptional activity

We have designed sequences which might have

Transcription Factor Binding Sites (TFBS)

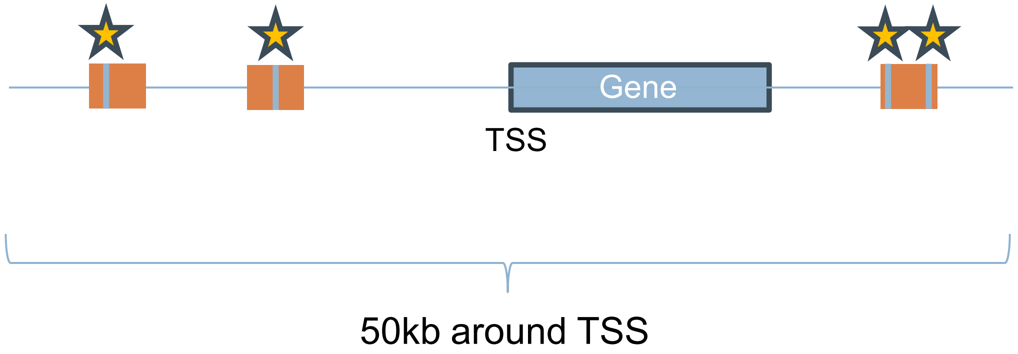
Short intro into the variant dataset used

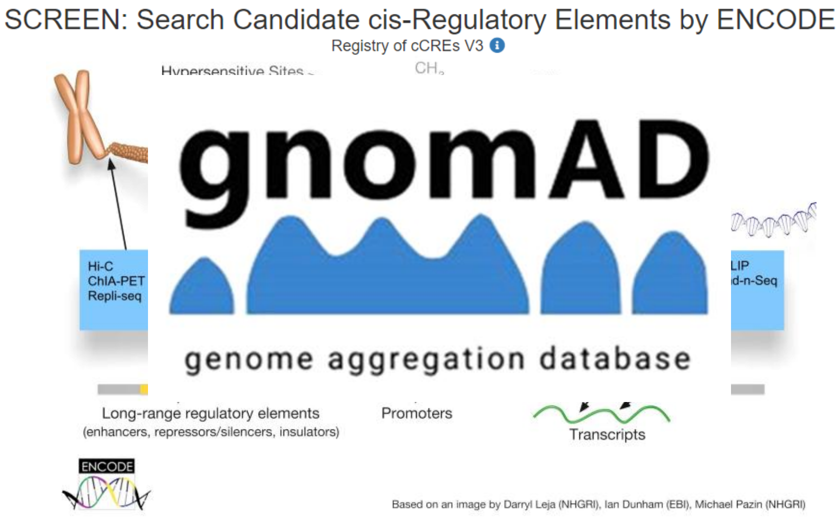

Theofilos Chalkiadakis


Mohan Dash


Max Schubach

Define candidate regulatory sequences



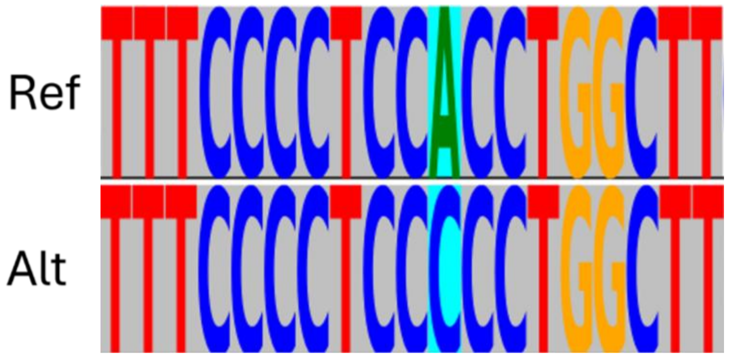


Known variation in cis-regulatory regions of a 50 kb window around disease associated genes

Motif Identification from Sequence

GOAL: Identify potential TFBS which binding affinity might be affected by the variant of interest

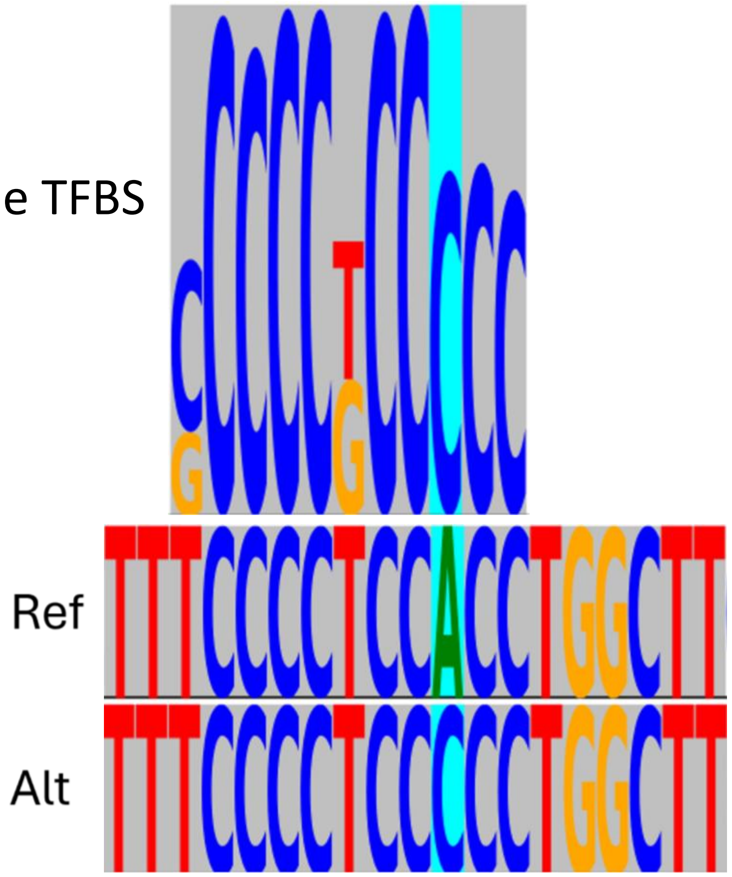
- We have reference and alternative allele
- We are interested in TFBSs



Motif Identification from Sequence

GOAL: Identify potential TFBS where binding affinity might be affected by the variant of interest

- We have reference and alternative allele
- We are interested in TFBSs
- The height of the character represents the „importance“ for the TFBS
- Let’s focus on the variant



Where does the TFBS fit best?

Is the motif fitting more likely the reference or alternative allele?

First: The C of the alternative allele increases the fit of the motif

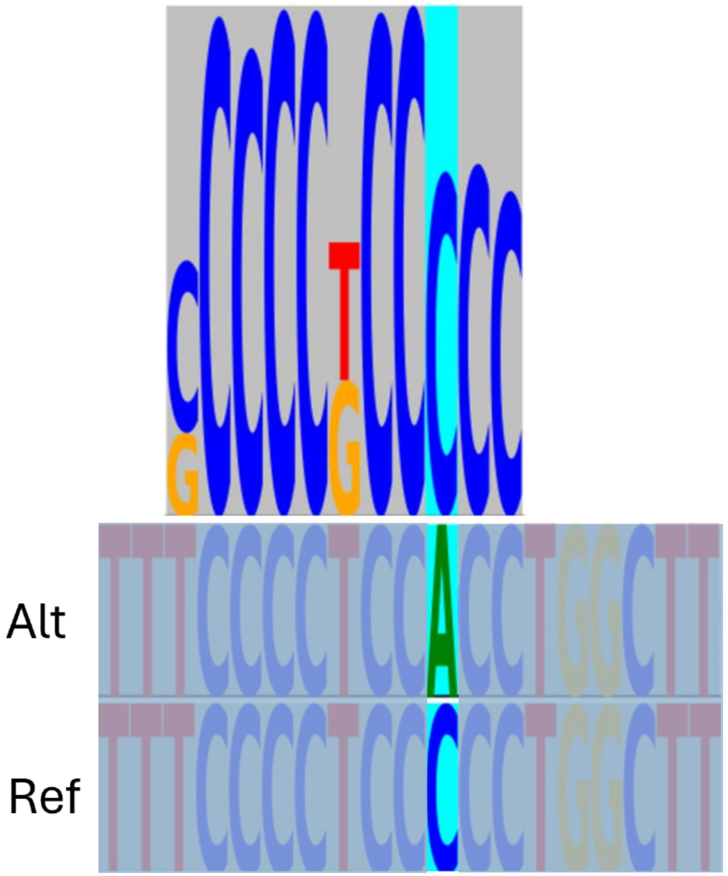
- This motif is predicted to be gained because of the variant

Second: The C of the reference allele increases the fitting of the motif

- This motif is predicted to be lost because of the variant

What do we need?

- Set of potential TFBS we are interested in

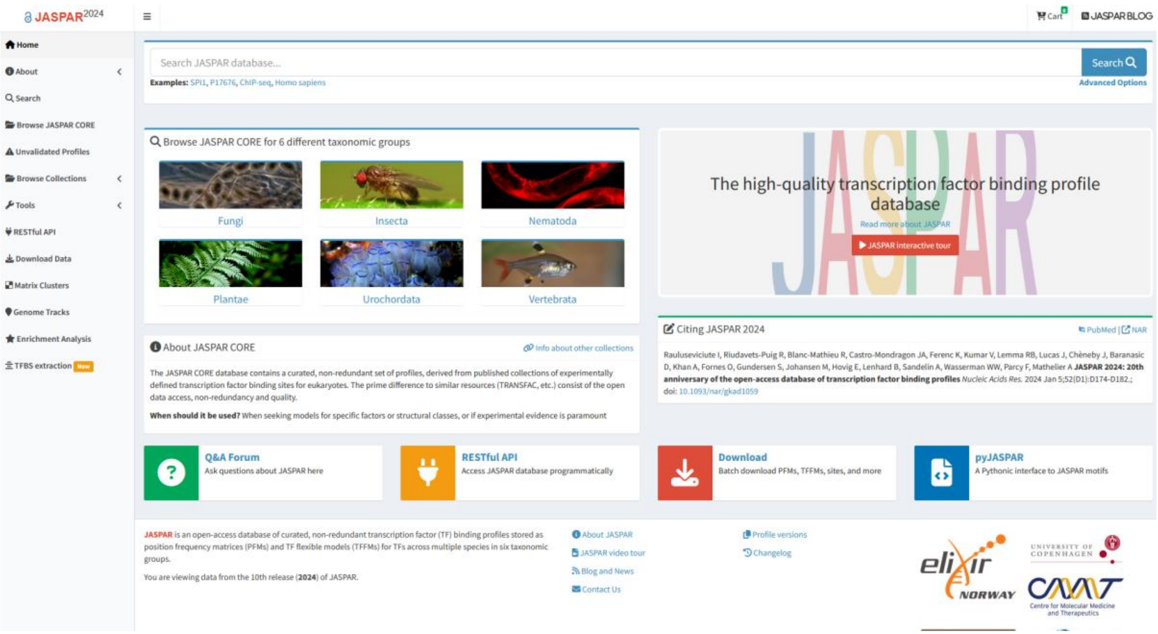


TFBS motif databases

Multiple options exist and depend on your task/data

Database	Focus / Scope	Source Organisms	Format	Notes
JASPAR	Non-redundant curated TF motifs	Primarily vertebrates, plants, insects	MEME, JASPAR	Open-access; manually curated; widely used in academia
HOCOMOCO	High-quality TF motifs from ChIP-seq	Human, mouse	MEME, TRANSFAC	High-quality, ChIP-seq-based motifs; ranking by quality
TRANSFAC	Comprehensive TF–DNA interactions	Multiple (mainly human)	Proprietary	Commercial license required; extensive annotations
CIS-BP	Predicted and experimental motifs	600+ species	MEME, custom	Combines multiple sources; includes inferred motifs
Factorbook	ENCODE-derived TF motifs	Human	JSON, BED	Motifs directly from ENCODE ChIP-seq data; ENCODE integration

TFBS motif databases: widely used in academia

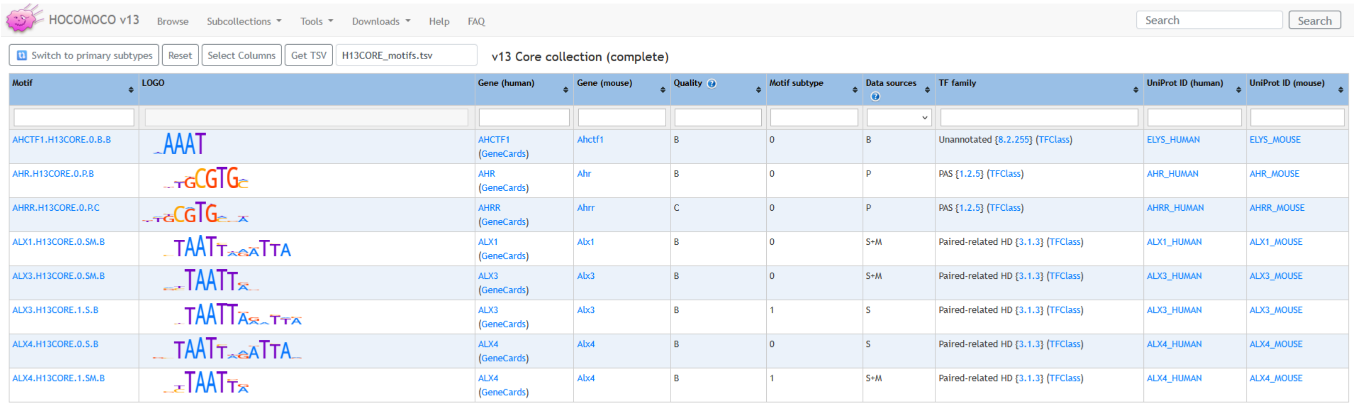


JASPAR

- Open access and manually curated motifs
- Broad cross-species coverage
- Compatibility with most tools (e.g., FIMO, HOMER)

HOCOMOCO

- Open access and manually curated motifs
- Human/mouse-focused, quality-ranked motifs
- Better for human clinical/genomics studies
- Compatibility with most tools (e.g., FIMO, HOMER)



Motif	LOGO	Gene (human)	Gene (mouse)	Quality	Motif subtype	Data sources	TF family	UniProt ID (human)	UniProt ID (mouse)
AHCTF1.H13CORE.0.B.B		AHCTF1 (GeneCards)	Ahctf1	B	0	B	Unannotated {8.2.255} (TFClass)	ELYS_HUMAN	ELYS_MOUSE
AHR.H13CORE.0.P.B		AHR (GeneCards)	Ahr	B	0	P	PAS {1.2.5} (TFClass)	AHR_HUMAN	AHR_MOUSE
AHRR.H13CORE.0.P.C		AHRR (GeneCards)	Ahrr	C	0	P	PAS {1.2.5} (TFClass)	AHRR_HUMAN	AHRR_MOUSE
ALX1.H13CORE.0.SM.B		ALX1 (GeneCards)	Alx1	B	0	S+M	Paired-related HD {3.1.3} (TFClass)	ALX1_HUMAN	ALX1_MOUSE
ALX3.H13CORE.0.SM.B		ALX3 (GeneCards)	Alx3	B	0	S+M	Paired-related HD {3.1.3} (TFClass)	ALX3_HUMAN	ALX3_MOUSE
ALX3.H13CORE.1.S.B		ALX3 (GeneCards)	Alx3	B	1	S	Paired-related HD {3.1.3} (TFClass)	ALX3_HUMAN	ALX3_MOUSE
ALX4.H13CORE.0.S.B		ALX4 (GeneCards)	Alx4	B	0	S	Paired-related HD {3.1.3} (TFClass)	ALX4_HUMAN	ALX4_MOUSE
ALX4.H13CORE.1.SM.B		ALX4 (GeneCards)	Alx4	B	1	S+M	Paired-related HD {3.1.3} (TFClass)	ALX4_HUMAN	ALX4_MOUSE

„Compatible with most tools“

Identifying given TFBS in a sequence set can be done using different options, again.

FIMO (Finding Individual Motif Occurrences)

- Scan sequences for motif matches
- Returns statistically significant motif hits (p/q-values)

HOMER (Hypergeometric Optimization of Motif EnRichment)

- Scan & discover (new) motifs
- Enrichment between two sets of sequences
- Less transparent statistical information about the resulting hits

We will just use one tool: FIMO

Input:

- Sequences as fasta and the motif database (HOCOMOCO v13)

Output:

- Table with motif id with a p-value and FDR corrected q-value (higher confidence = lower)

Column	Description
motif_id	ID of the matched motif
sequence_name	Name of the sequence with the motif
start/stop	Start/stop positions of the motif match
strand	Strand (+/-)
matched_sequence	Sequence matching the motif
p-value	Motif match p-value (default threshold: 1e-4)
q-value	FDR-corrected p-value (higher confidence = lower)

Combine FIMO Output with Variant Information

Variant information per row:

- reference and alternative allele (name and sequence), variant position, measured variant effect (log2(alt – ref)) and an adjusted p-value

Small example:

motif_id	start	stop	variant_pos	ALT_q-value	REF_q-value
ATMIN.H13CORE.0.P.B	62	82	63	0.0107	0.00643
RXRA.H13CORE.3.P.B	43	63	48	0.0832	0.0988
SP5.H13CORE.1.P.B	67	91	81	NA	0.0165
TTF1.H13CORE.0.PSG.A	106	126	113	0.182	NA
ANDR.H13CORE.0.P.B	186	202	63	0.187	0.187

Focusing on variant overlapping TFBS

FIMO output: start and end of the motifs found in the sequence

Variant info: variant position (all variants in this example set are significant)

Looking into the resulting table:

- 513 variant overlapping TFBS predicted
- 13 different variants with at least one predicted variant overlapping TFBS

Huge number and potentially many false positives

How confident are we about these significant hits? How can we improve confidence?

Improve confidence in the TFBS predictions

Problem: Motifs are short sequences => many chance matches (false positives)

How to prioritize?

- Do you remember the q-value? We can be more stringent with the q-value
- We could focus on certain TFs we are interested in
- Or even use additional experimental data (e.g. ChIP-seq)

q-value threshold of 0.1

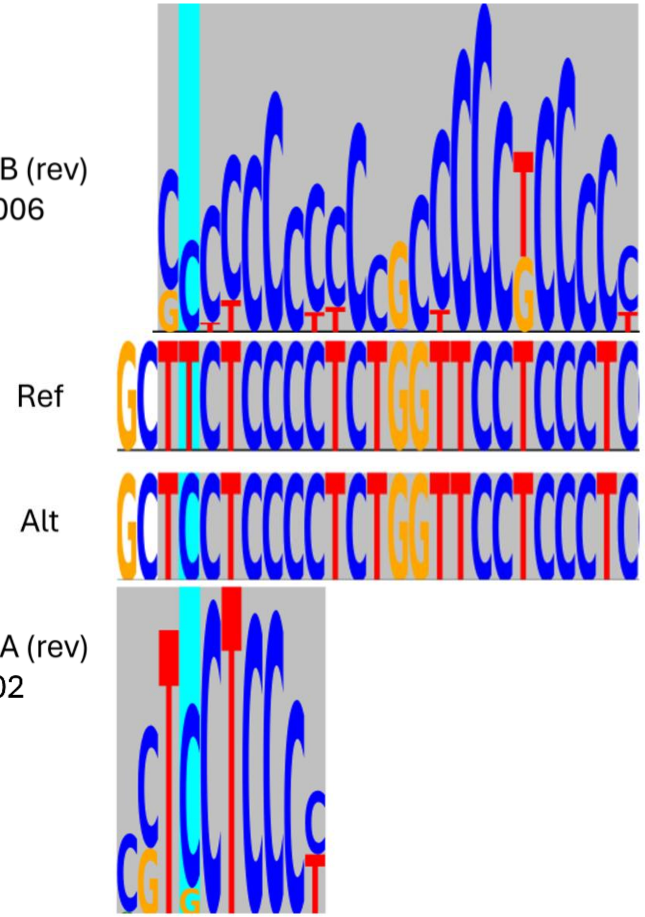
- 383 variant overlapping TFBS predicted
- 13 different variants

Does this do what we expect?

Which TFBS fits better? Eye vs q-value

Another possibility: multiple TFBS overlapping the same variant => Which fits better?

- Does the TFBS at the bottom fit well?
- Would you say the motif at the top is fitting better?
- Top motif: q-value from fimo is 0.006
- Should the q-value of the bottom motif be higher or lower?



Result:

- So FIMO predicts the top motif with higher confiden

Reason:

- Motif length is a bias for the computed statistic
- Shorter motifs are more likely to be found just by chance

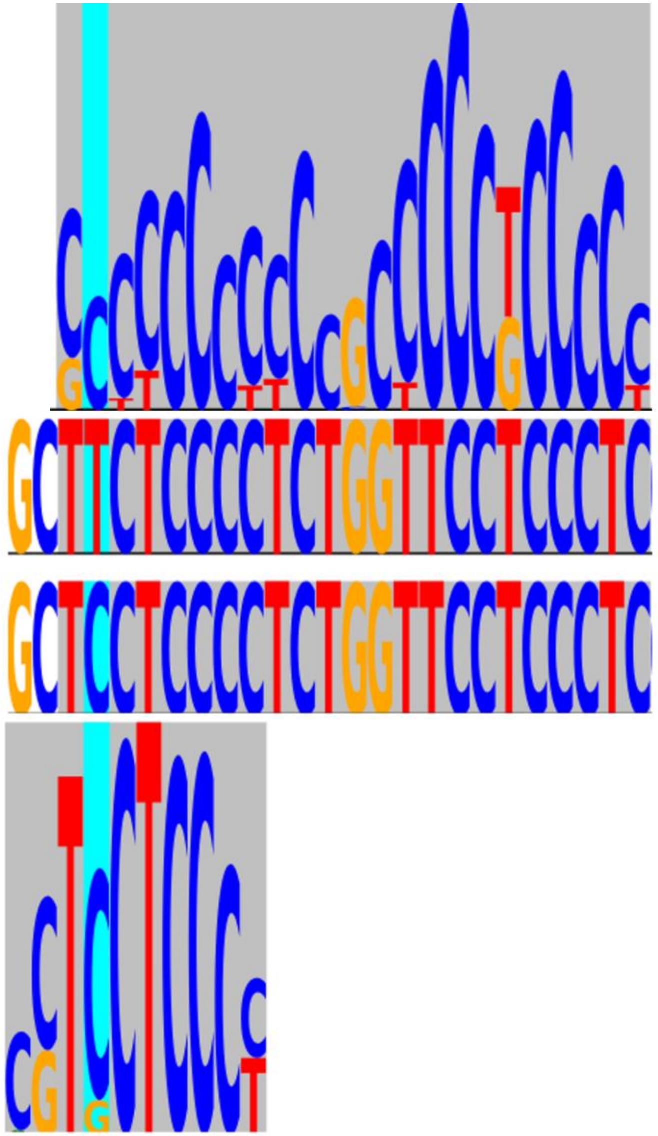
Which TFBS fits better? Eye vs q-value

PATZ1.H13CORE.0.P.B (rev)
Fimo q-value (Alt): 0.006

Ref

Alt

ZN263.H13CORE.0.PM.A (rev)
Fimo q-value (Alt): 0.0602



New measure: how well fits the motif the variant location?

Score the motif fitting

for a unified size around the variant position

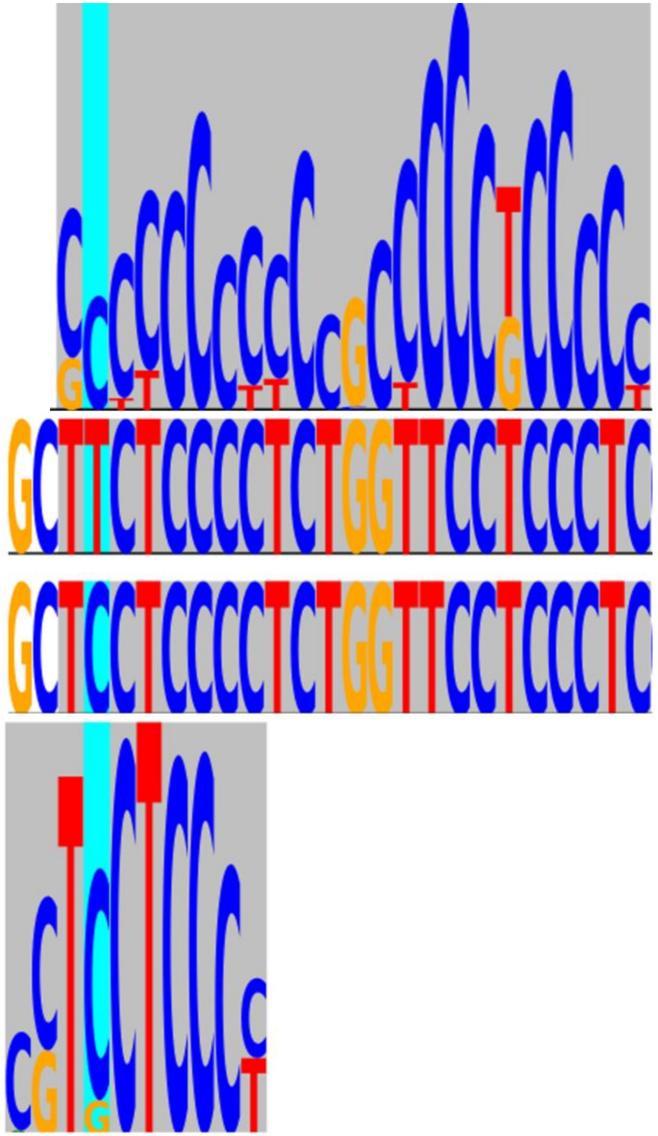
PATZ1.H13CORE.0.P.B (rev)
Fimo q-value (Alt): 0.0056 8.05

Filtering for well fitting TFBS only:

- E.g. max bitscore >=8
- 51 TFBS predictions for 12 variants le

Ref

Alt



ZN263.H13CORE.0.PM.A (rev)
Fimo q-value (Alt): 0.0092 8.39

Use additional data to increase confidence of the predictions

Imagine you have a binarized table and a gene is active in many different cell-types:

gene_symbol	Adipocytes	Fibroblasts	Excitatory neurons	Inhibitory neurons
CTCF	1	1	1	1
SP5	0	1	0	1
SOX4	1	1	1	1
ASPDH	0	0	1	1
GATA4	0	1	0	0

Let's focus on TFBS which are expressed either in excitatory or inhibitory neurons

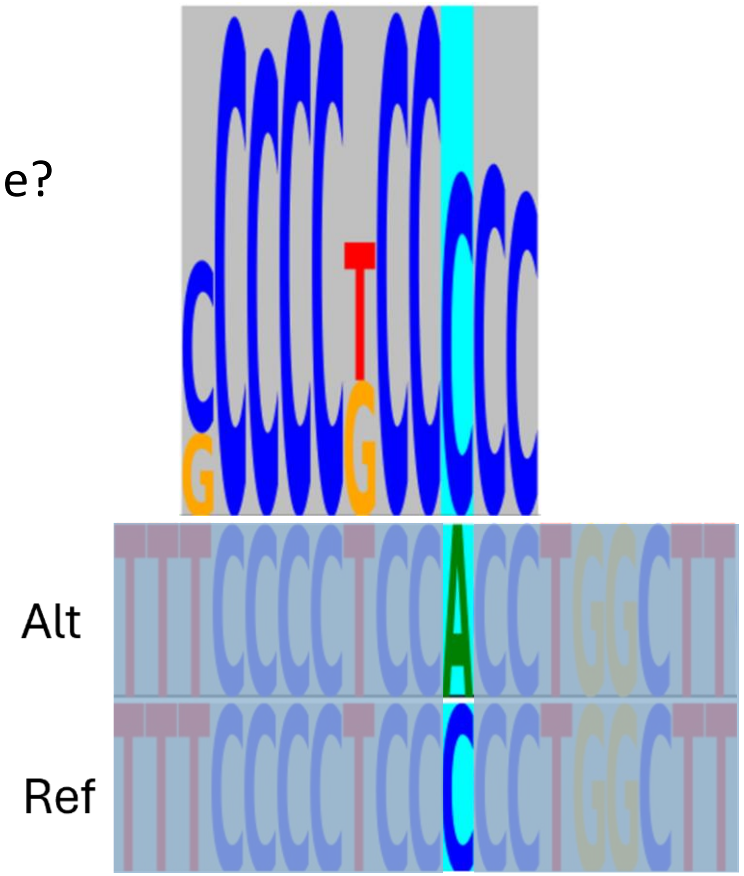
- 26 TFBS predictions left

Is a predicted TFBS likely gained or lost?

A TFBS can be found either on both reference and alternative allele or only in one of them, depending on the result from FIMO

The question can be rephrased to:

- Is the motif fitting more likely the reference or alternative allele?



How many potential TFBS were gained and lost?

TFBS change	Number of motifs	Number of variants
gain	12	8
loss	10	8
No change	4	3

Interpret variant effect on TFBS level OR Combine potential motif gain/loss and measured variant effect

Activating TFBS:

- if ALT binds better and positive effect => motif is predicted to increase transcription in this cell type
- if REF binds better and negative effect => motif is predicted to increase transcription in this cell type

Repressing TFBS:

- if ALT binds better and negative effect => motif is predicted to decrease transcription in this cell type
- if REF binds better and positive effect => motif is predicted to decrease transcription in this cell type

TFBS interpretation	count
Predicted transcriptional increasing TFBS	12
Predicted transcriptional decreasing TFBS	10

You might ask: is a gain of a TFBS always transcription increasing?

TFBS change	Number of motifs
gain	12
loss	10
No change	4

TFBS interpretation	count
Predicted transcriptional increasing TFBS	12
Predicted transcriptional decreasing TFBS	10

If we group by the TFBS interpretation, we can see that this is not true

TFBS interpretation	Gain	Loss
Predicted transcriptional decreasing TFBS	4	6
Predicted transcriptional increasing TFBS	8	4

Conclusion

Do not trust the results of TFBS predictions without sanity checking

Being more stringent might reduce the number of false positives

So does using additional information

- Focusing on a reasonable subset of TFs
- Focusing on only the well-fitting motifs

Obviously, a gained motif does not need to increase the transcriptional activity

Interpretations of these motifs do not proof actual binding

- Functional validation is the next step

Discussing data limitations

ISMB-ECCB 2025: *Tutorial IP2 (MPRAs and IGVF Resource)*

Arjun Devadas, Kilian Salomon, Martin Kircher, Max Schubach
Regulatory Genomics @ UzL/UKSH, Computational Genome Biology @ BIH

July 20, 2025

This session is an opportunity to discuss the boundaries of current MPRA technology and modeling approaches and identify areas where more research or methodological development is needed.

- Share Your Challenges: Can you share specific examples from your own research where you encountered limitations related to variant effects, sequence context, or cell-type specificity?
- Interpretation of MPRA Data: What do MPRA limitations mean for drawing robust conclusions from MPRA experiments?
- Implications for Predictive Modeling: How do data limitations impact the accuracy and generalizability of models trained on MPRA data, and how these models can be improved to account for cell-type specificity or complex sequence contexts?
- Importance of the genomic context: MPRA activities are often tested outside their native genomic context and are difficult to translate into phenotypic or organismal effects. What are strategies to better connect in-vitro MPRA findings to a broader biological and clinical relevance?
- Design of future MPRAs: How do current limitations inform the design of future MPRA experiments or computational approaches to overcome current hurdles, such as expanding cell-type diversity or improving sensitivity to subtle cell-type specific effects?

Thank You! Questions?

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BIH Berlin Institute of Health @Charité

Participant questions and feedback

ISMB-ECCB 2025: *Tutorial IP2 (MPRAs and IGVF Resource)*

Arjun, Kilian, Martin & Max
Regulatory Genomics @ UzL/UKSH, Computational Genome Biology @ BIH

July 20, 2025

UNIVERSITÄT ZU LÜBECK

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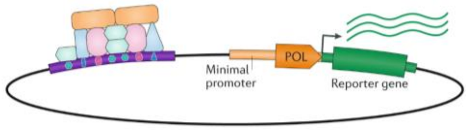
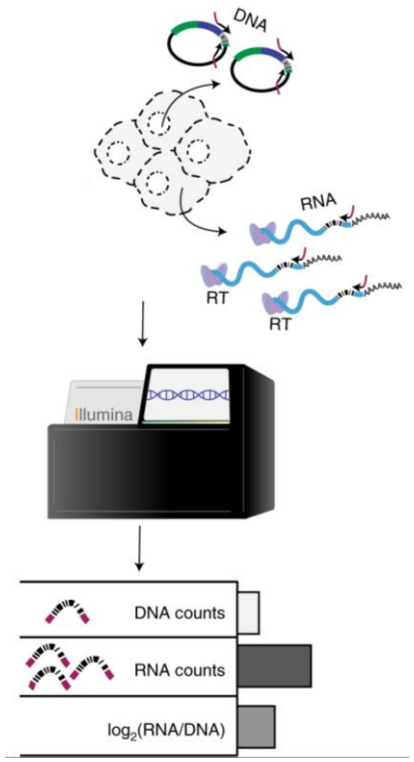
Berlin Institute of Health @Charité

Revisiting our Learning Objectives

- Introduction to IGVF and how to access its data.
- MPRA Experiments: Explanation of experimental design, including the association of barcode/tag sequences to tested regulatory sequences and variants.
- Step-by-step data processing using MPRAsnakeflow, including construction of count tables and interpretation of quality control metrics.
- Statistical analysis of DNA and RNA barcode counts using BCalm.
- Applications of MPRA data like identifying active regulatory regions and significant variant effects, optimising cell-type-specific expression through synthetic sequences.
- Sequence-based modeling of regulatory sequence activity, i.e. using deep learning models to predict open chromatin and MPRA activity, investigate transcription factor motif importance and composition.



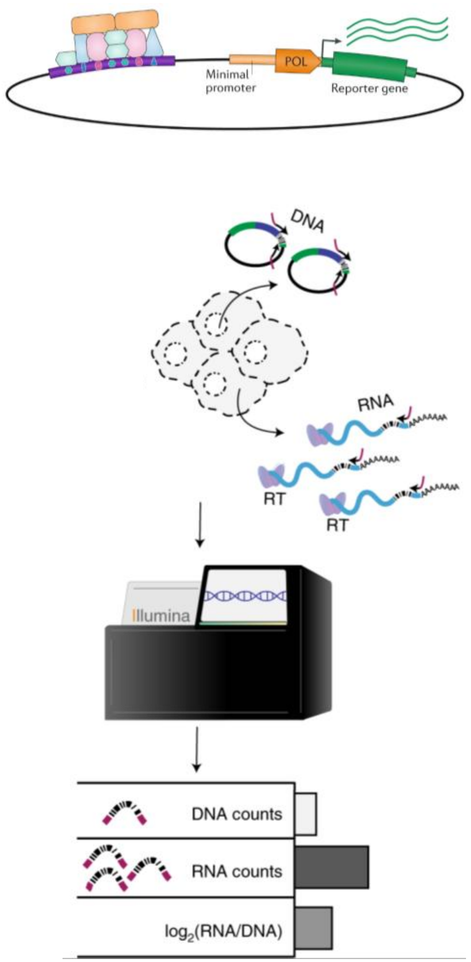


Questions?

Questions for you? (1)

Reflecting on your motivation for attending this tutorial, **did the content covered** – from the introduction to IGVF and MPRA experiments to data processing, analysis, and applications – **meet your expectations?**

Which specific learning objective, such as understanding MPRA experiments, mastering data processing steps with MPRAsnakeflow, or exploring applications like variant identification or sequence modeling**, did you find most valuable for your current or future research?**

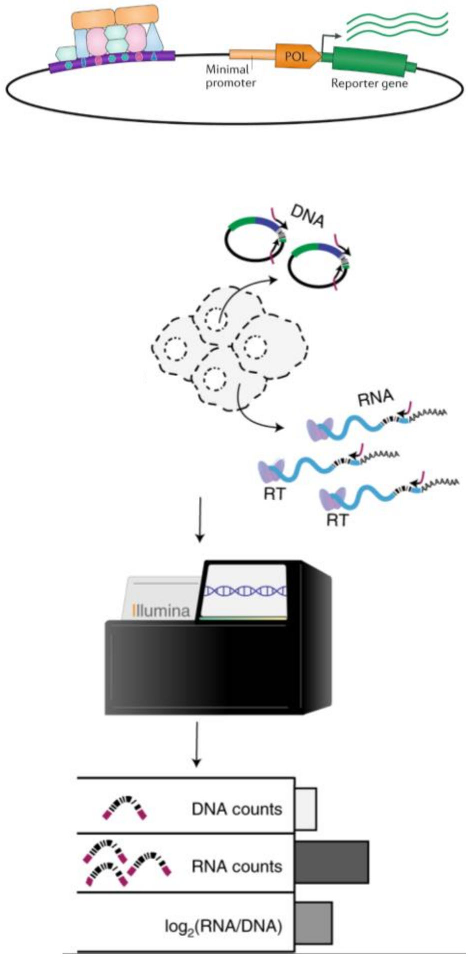


Questions for you? (2)

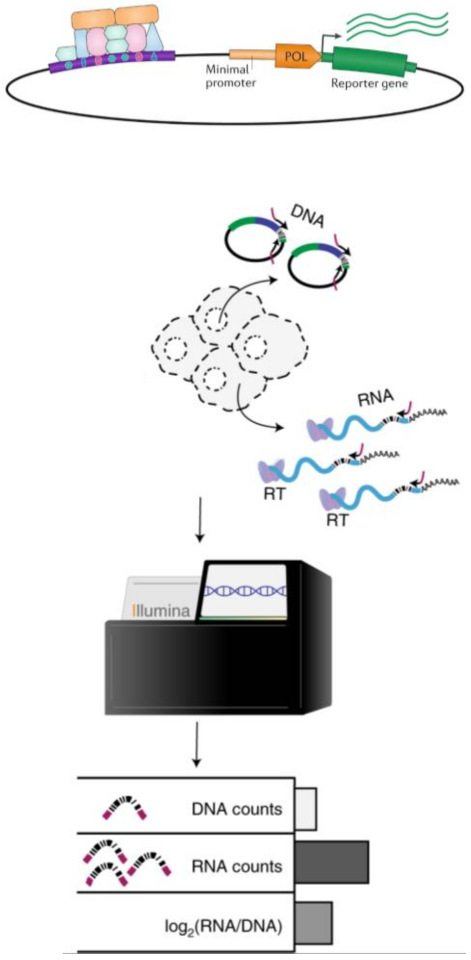
What part of the tutorial content or hands-on practice do you believe will have the most immediate impact on your research in functional genomics, variant analysis, or sequence model training?

How confident do you feel about starting to process and analyze MPRA data using tools like MPRAsnakeflow or BCalm after participating in the hands-on exercises on Google Colab?

What further support might help increase your confidence?



Questions for you? (3)



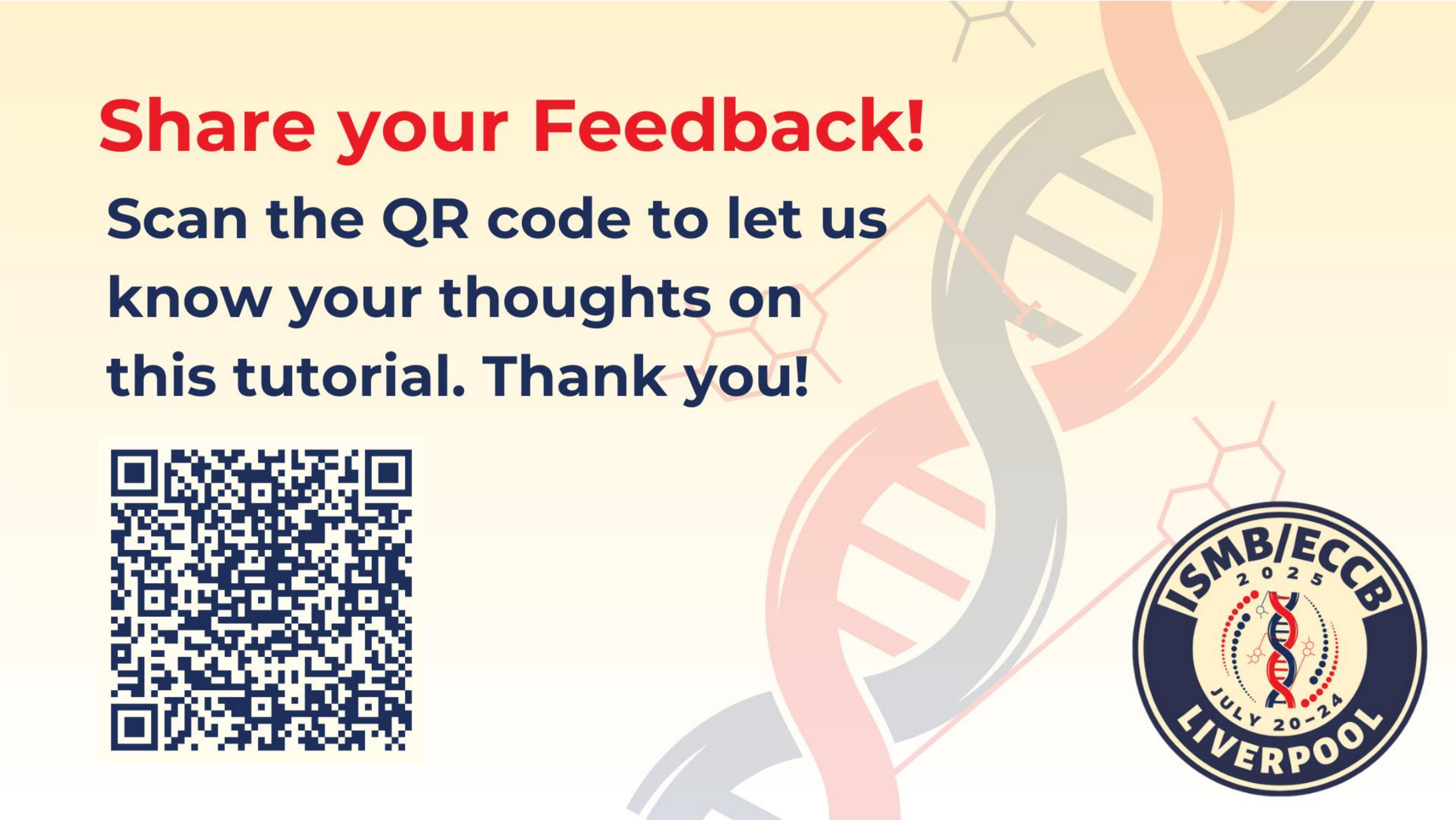
How useful was the introduction to the IGVF consortium and learning how to access its data resources?

Do you see opportunities to leverage IGVF data in your work?

Thanks to everyone to contributing and making this a successful day!

- **Thank you to all participants for their active role!**
- **Thank you to our speakers:**
  - Arjun Devadas, University of Lübeck / UKSH
  - Kilian Salomon, Berlin Institute of Health at Charité
  - Martin Kircher, University of Lübeck / UKSH & Berlin Institute of Health at Charité
  - Max Schubach, Berlin Institute of Health at Charité
- **And everyone supporting the tutorial preparation:**
  - Michael Love, University of North Carolina
  - Jonathan Rosen, University of North Carolina
  - Pia Keukeleire, University of Lübeck / UKSH
  - Pyaree-Mohan Dash, Berlin Institute of Health at Charité
  - **ISMB-ECCB reviewers and Tutorial Organizers**

Please participate in the ISMB-ECCB tutorial feedback!



Feel free to reach out!

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